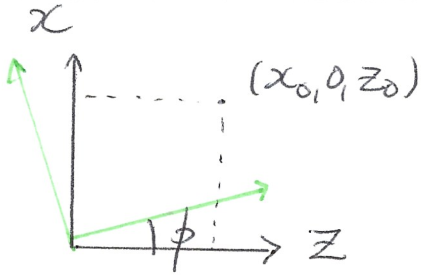
Rond y-as :



rotatie zoals in 2D maar in andere richting $\phi \rightsquigarrow -\phi$

$\left(\begin{array}{l}\cos(-\phi) = \cos\phi \\ \sin(-\phi) = -\sin\phi\end{array}\right)$

$$(x_1, y_1, z_1) = (x_0, y_0, z_0) \begin{pmatrix} \cos\phi & 0 & \underline{\sin\phi} \\ 0 & 1 & 0 \\ \underline{-\sin\phi} & 0 & \cos\phi \end{pmatrix}$$

Sequentie van rotaties & translaties :

① Translatie in matrixvorm? $(x_1, y_1, z_1) = (x_0, y_0, z_0) \cdot$? matrix 3x3 onmogelijk

$$\text{zodat} \begin{cases} x_1 = x_0 + a \\ y_1 = y_0 + b \\ z_1 = z_0 + c \end{cases}$$

Oplossing: homogene coördinaten : $(x, y, z, 1)$

dan $$(x_1, y_1, z_1, 1) = (x_0, y_0, z_0, 1) \underbrace{\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline a & b & c & 1 \end{array}\right)}_{\text{translatiematrix}}$$

② 4x4 rotatiematrices :

bv $$M_Z(\phi) = \left(\begin{array}{ccc|c} \cos\phi & -\sin\phi & 0 & 0 \\ \sin\phi & \cos\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right)$$